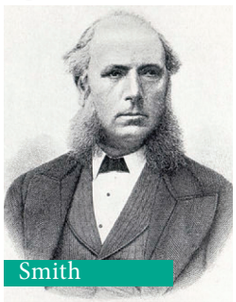
The seven LEDs are labelled *A* through *G*. By forward biasing different LEDs, we can display the digits 0 to 9. For example, if LEDs *A,B,C, D and G* are lit (by forward biasing them), then the display will show the number 3.

There are two types of seven segment LED arrangement, they are

- Common Anode Type
- Common Cathode Type

6.3. LIGHT DEPENDENT RESISTOR (LDR)

A light dependent resistor is a semi-conductor device, whose resistance value changes when illuminated (exposed) with light. It is also called as photo-resistor or photo-conductive cell.


Smith

History: LDR is a photoconductive device made from Selenium. It was discovered by Smith in 1873. In 1952, Rollin and Simmons developed their photoconductors using Silicon and Germanium.

Construction: The Figure 6.3(a, b) shows the basic structure and symbol of LDR. It consists of a thin layer of cadmium sulphide (*CdS*), which is used as photo cell. It also contains Antimony or Indium which could be deposited on an insulating substrate. Ohmic contacts are made at the opposite ends of the *CdS* layer. The value of resistance is about 2 MΩ, at absolute darkness. But, strong incident light focussing on it reduces the resistance value fall below the 100 Ω.

Working Principle

When radiation is incident on a semiconductor, amount of light is absorbed by the material and its conductivity increases. This effect is called photo-conductive effect.

The radiation falls on the semiconductor produces charge carriers. This charge carrier produces electron-hole pairs. If the intensity of the light falling on the surface increases, it will generates large number of electron hole pairs. This will lower the value of resistance of the photo-conductor.

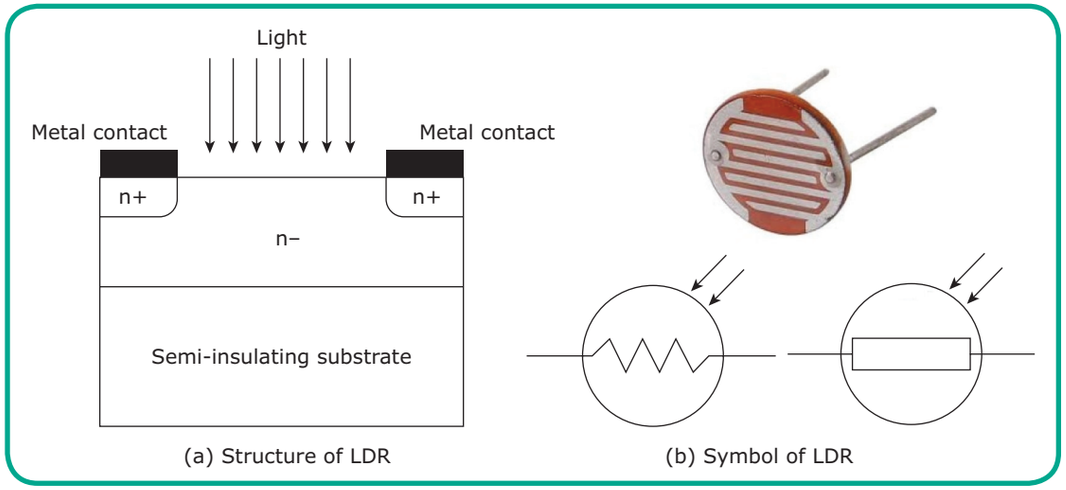


Figure 6.3 Light Dependent Resistor

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